Table 1: Results for CG, GS, HP-C, and HP-KR using the 20th percentile versus the 66th percentile layer activations. We compare AUROC and RECALL across all probe families using `Mistral-Small-3.1-24B`.

Method	CG		GS		HP-C		HP-KR		Avg
	L20	L66	L20	L66	L20	L66	L20	L66	
AUROC									
Logistic	0.76	**0.84**	0.40	**0.56**	**0.64**	0.31	**0.14**	0.13	0.47
TPC	0.66	**0.67**	**0.47**	0.38	**0.52**	0.38	**0.16**	0.12	0.42
Truth2D	0.73	**0.80**	0.41	**0.61**	**0.64**	0.32	0.14	**0.15**	**0.48**
Mass-Mean	**0.62**	0.44	0.44	**0.50**	**0.62**	0.31	0.14	**0.19**	0.41
INLP	0.74	**0.82**	0.41	**0.55**	**0.64**	0.31	0.14	**0.15**	0.47
Follow-Up	0.54	**0.55**	**0.43**	0.29	**0.47**	0.44	**0.42**	0.12	0.41
Upper-Bound	1.00	**1.00**	**1.00**	1.00	0.75	**0.82**	0.94	**0.95**	0.93
Average	0.72	**0.73**	0.51	**0.56**	**0.61**	0.41	**0.30**	0.26	0.51
RECALL									
Logistic	0.91	**0.99**	0.08	**0.10**	**0.96**	0.76	**0.01**	0.00	**0.48**
TPC	**0.91**	0.76	**0.03**	0.01	**0.76**	0.66	**0.01**	**0.01**	0.39
Truth2D	0.86	**0.98**	**0.06**	0.00	0.39	**0.66**	**0.01**	0.00	0.37
Mass-Mean	**0.97**	0.90	**0.05**	0.00	**0.00**	**0.00**	**0.01**	0.00	0.24
INLP	0.93	**0.99**	**0.08**	0.01	**0.96**	0.65	**0.01**	0.00	0.45
Follow-Up	**1.00**	0.81	**0.65**	0.34	**0.14**	0.04	**0.12**	0.02	0.39
Upper-Bound	0.93	**1.00**	**1.00**	**1.00**	0.52	**0.66**	**0.92**	0.81	0.85
Average	**0.93**	0.92	**0.28**	0.21	**0.53**	0.49	**0.16**	0.12	0.45

depths: an early 20th-percentile baseline and a deeper 66th-percentile layer.

For the sparsity analysis (Table 2), we establish a strictly controlled comparison by utilizing `Gemma-2-27B` for *both* the dense and sparse pipelines. This guarantees that any observed performance differences arise solely from the feature representation rather than a mismatch in the underlying foundation model. We extract both the raw dense hidden states and the corresponding sparse SAE features (using the `gemma-scope-27b-pt-res-canonical` weights with a hidden dimension of 131,072) from Layer 31. Because `Gemma-2-27B` features exactly 46 layers, this target corresponds to approximately 67.4% depth, aligning with the peaked middle-layer activation pattern observed in Boxo et al. [3].

Across all setups, training utilizes 1,000 DolusChat samples, a learning rate of 10^{-3}, 20 probe epochs, and a batch size of 64.

4.2 Evaluation Metrics

To assess the detectability of deception across different representations, we report the Area Under the Receiver Operating Characteristic Curve (AUROC) and standard Recall (evaluated at a default 0.5 probability threshold). AUROC serves as our primary metric, because it provides a threshold-independent measure of the global linear separability between truthful and deceptive internal states. Rather than optimizing for a specific operational threshold, this approach allows us to answer our core mechanistic question: to what extent do dense versus sparse representations inherently encode the structural typology of a lie?

We acknowledge that in real-world operational deployments, lie detectors require strict False Positive Rate (FPR) constraints or precision-focused tuning, as over-predicting deception can falsely penalize honest model behavior. Because our primary objective is to evaluate fundamental representation geometries rather than to deploy a production-ready classifier, we leave the exact mapping of FPR trade-offs at strict operational thresholds to future work.

5 Results and Analysis

In this section, we present the empirical findings from our probing experiments. To systematically unpack the mechanics of deception detection in large language models, we structure our analysis around the four core factors hypothesized in Section 1: representation depth, probe expressivity, sparsity, and lie typology. For each dimension, we present our quantitative evaluations followed by an interpretation of the underlying mechanisms driving these results.

5.1 The Effect of Representation Depth

To investigate whether deceptive representations mature in the later stages of the model, we extracted dense hidden states at two distinct depths: an early layer (the 20th percentile) and a deeper layer (the 66th percentile). As shown in Table 1, the effect of representation depth is highly dataset-dependent rather than universally monotonic.

On the self-referential datasets (CG and GS), the 66th percentile layer consistently improves AUROC across most probes. This suggests that for role-based or secret-keeping deception, deeper representations are needed to capture the internal contradiction between the model's assigned persona and its underlying knowledge.

Table 2: A strictly controlled comparison of dense hidden states and sparse SAE features extracted from the exact same foundation model (`Gemma-2-27B`) at 66th percentile layer. By evaluating both representation types on the identical underlying model, we isolate the specific effect of sparsity on detectability. Both variants were trained on the full DolusChat mixture and evaluated across CG, GS, HP-C, and HP-KR.

Method	CG		GS		HP-C		HP-KR		Avg
	Dense	SAE	Dense	SAE	Dense	SAE	Dense	SAE	
AUROC									
Logistic	**0.67**	**0.67**	**0.51**	0.46	**0.48**	0.48	**0.35**	0.31	0.49
TPC	**0.69**	0.66	**0.48**	0.43	0.51	**0.52**	**0.54**	0.48	**0.54**
Truth2D	**0.66**	0.65	**0.46**	0.44	**0.48**	0.47	**0.38**	0.34	0.49
Mass-Mean	0.53	**0.63**	0.44	**0.45**	0.48	**0.49**	**0.37**	0.34	0.47
INLP	0.67	**0.68**	**0.50**	0.45	**0.48**	0.47	**0.36**	0.30	0.49
Follow-Up	**0.48**	0.46	**0.58**	0.56	0.53	**0.53**	0.43	**0.41**	0.50
Upper-Bound	**0.92**	**0.92**	0.77	**0.78**	**0.61**	**0.61**	**0.79**	0.76	0.77
Average	0.66	**0.67**	**0.53**	0.51	**0.51**	**0.51**	**0.46**	0.42	0.53
RECALL									
Logistic	**0.93**	0.86	**0.22**	0.11	**0.32**	0.29	**0.32**	0.13	0.40
TPC	**0.84**	0.78	0.20	**0.16**	0.38	**0.40**	0.20	**0.36**	**0.42**
Truth2D	**0.90**	0.87	0.11	**0.11**	**0.31**	**0.31**	**0.24**	0.13	0.37
Mass-Mean	0.19	**0.66**	**0.37**	0.18	**0.47**	0.41	**0.46**	0.28	0.38
INLP	**0.92**	0.79	**0.12**	0.03	**0.28**	0.23	**0.22**	0.10	0.34
Follow-Up	0.57	**0.61**	0.47	**0.48**	**0.20**	0.19	0.32	**0.38**	0.40
Upper-Bound	0.82	**0.84**	0.70	**0.77**	**0.40**	0.32	**0.59**	0.54	0.62
Average	0.74	**0.77**	**0.31**	0.26	**0.34**	0.31	**0.33**	0.27	0.42

Conversely, the harm-pressure datasets (HP-C and HP-KR) exhibit stronger separability at the 20th percentile layer. This early-layer advantage possibly occurs because deceptive responses in these datasets are heavily entangled with safety protocols, refusal tendencies, and harmful-context cues. In these cases, deeper layers may over-abstract the representation, washing out surface-level features that linear probes rely on to detect the lie.

Furthermore, the recall metrics reveal a distinct split depending on the dataset category. On the harm-pressure datasets, the 20th percentile layer provides a more robust and stable decision boundary at the default operating threshold. Conversely, for the baseline logistic probe, the 66th percentile layer yields higher recall on the self-referential datasets (CG and GS). This reinforces the broader conclusion that the optimal representation depth is strictly tied to the nature of the deceptive context rather than a universal property of the model.

A striking anomaly in Table 1 is the sub-random AUROC performance (e.g., 0.12–0.16) on the HP-KR dataset across nearly all probes. It is important to note that this is not an erroneous result; the original Liars' Bench benchmark also reports substantially sub-chance AUROC results in this setting (for example, finding an AUROC of 0.12 for Llama 3.3 70B with a mean probe on HP-KR) [12]. Because an AUROC significantly below 0.50 indicates systematic anti-correlation, this is not merely a failure to generalize, but an active *anti-transfer*. The probes trained on direct, context-free lies in DolusChat appear to latch onto a feature direction that maps inversely to the safety-driven, self-knowledge deception required in HP-KR. This is a crucial mechanistic insight: the internal representation of a direct fabrication is inverted compared to the representation of a model falsely denying hazardous knowledge under safety pressure.

5.2 Impact of Probe Expressivity

To investigate whether more expressive probe families—those capturing feature interactions or subspace structure—outperform standard logistic regression, we compare logistic regression against TPC, Truth2D, Mass-Mean, and INLP. We treat the follow-up and upper-bound rows separately, since they serve as prompt-based and ceiling baselines rather than expressivity-oriented methods.

As shown in Table 1, the empirical results provide mixed support for the expressivity hypothesis. Truth2D provides the clearest gains among the structured probes, particularly on GS, where it slightly improves AUROC over the logistic regression baseline. This is consistent with the idea that explicitly disentangling truth from polarity can create a more robust deception subspace. Conversely, TPC does not yield a consistent improvement over logistic regression, suggesting that low-order polynomial interactions are either insufficient for this task or too sensitive to the training distribution to generalize reliably. Mass-Mean is generally the weakest of the structured probes, indicating that this simplified geometric summary does not preserve enough discriminative information.

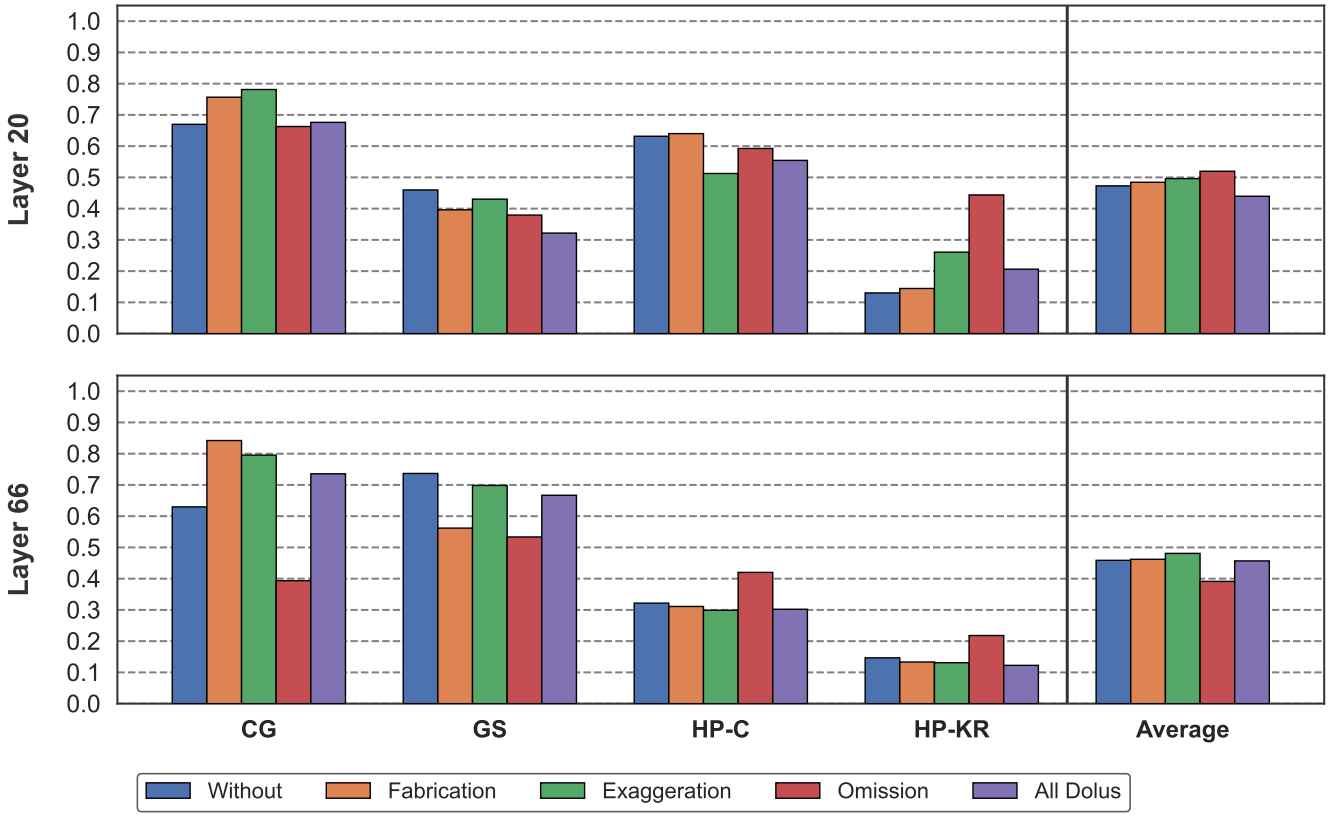


Figure 2: Comparison of logistic-probe AUROC across lie-typology training conditions and evaluation datasets. We compare probes trained without DolusChat, on fabrication-only DolusChat, on exaggeration-only DolusChat, on omission-only DolusChat, and on the full DolusChat mixture, at 20th-percentile and 66th-percentile.

Crucially, while INLP serves as a useful geometric baseline, its performance should be interpreted as a measure of subspace robustness rather than standard detection. As shown in Table 1, INLP is often close to logistic regression, but not identical across all datasets and depths: on CG it reaches 0.74 and 0.82 AUROC versus 0.76 and 0.84 for logistic regression, on GS it reaches 0.41 and 0.55 versus 0.40 and 0.56, on HP-C it matches the baseline at 0.64 and 0.31, and on HP-KR both methods remain near chance. These results suggest that iteratively projecting out linear directions removes some information, but does not uniformly collapse the deception signal. Rather than indicating that deception is confined to a single low-dimensional subspace, the findings are more consistent with a partly distributed representation that is only partially removed by linear null-space projection.

5.3 Impact of Sparse SAE Features

Unlike the earlier fabrication-only comparison, this experiment uses SAE results computed across all DolusChat lie categories. Furthermore, due to computational constraints, SAE features were only extracted at 66th percentile layer. To keep the comparison matched, we therefore evaluate the dense branch on the same all-category DolusChat subset and at the same layer, so any differences reflect the representation type rather than changes in lie-type coverage or depth.

To examine whether sparse SAE features preserve or improve deception separability relative to dense hidden states, we compare Gemma-2-27B hidden activations against Gemma-2-27B SAE features at 66th percentile layer in Table 2.

The results presented in the table suggest that, within this specific experimental setup, the transition from dense hidden states to sparse SAE features does not appear to yield a clear improvement in deception detection performance. While the sparse representations maintain a relative functional parity with the dense branch, they generally achieve slightly lower or equivalent average scores across most datasets and probing methods. This suggests that the SAE reconstruction at this depth preserves existing deception-related signals rather than enhancing their separability.

The main exception is the harm-pressure setting, where SAE features improve some individual probe scores and appear to help disentangle deception from overlapping safety-related signals. However, this effect is not uniform across all methods or datasets, and it does not translate into a consistent overall gain. The upper-bound rows further support this interpretation: dense and sparse features

remain close in performance, but the dense branch is still slightly stronger overall.

5.4 Impact of Lie Typology

To test whether the structural category of a lie affects detectability, we compare AUROC across five training conditions: without DolusChat, fabrication-only, exaggeration-only, omission-only, and the full DolusChat mixture. We restrict this analysis to the baseline logistic regression probe because, as established in Section 5.2, it provides the most stable and representative baseline across datasets. By standardizing on this robust linear boundary, we successfully isolate the effect of the training distribution on linear separability without compounding variables from fluctuating probe expressivity. Figure 2 summarizes these results at the 20th percentile layer and the 66th percentile layer.

The bar plot shows that lie typology matters, but not in a simple universal order. At the 20th percentile, the best-performing condition depends on the dataset: exaggeration is strongest on CG, the no-DolusChat baseline is strongest on GS, fabrication is strongest on HP-C, and omission is strongest on HP-KR. Averaged across datasets, omission has the highest AUROC at the 20th percentile layer, largely because of its strong performance on HP-KR. This suggests that omission is the most recoverable training condition at the shallower layer, but only in the harm-pressure setting.

At the 66th percentile layer, the pattern shifts. Fabrication becomes strongest on CG, the no-DolusChat baseline remains strongest on GS, and omission becomes the strongest condition on both HP-C and HP-KR. On average, exaggeration achieves the highest AUROC at the 66th percentile layer, while omission drops substantially relative to its 20th percentile layer performance. This indicates that deeper representations sharpen some lie-typology signals while suppressing others, especially when the lie depends on contextual withholding rather than direct contradiction.

The full DolusChat mixture is never the best-performing condition and is weakest at the 20th percentile layer, with a near-bottom average at the 66th percentile layer as well. This suggests that combining lie types does not automatically improve linear separability and may instead blur the deception signal. Overall, the figure supports a qualified conclusion: lie typology does affect detectability, but there is no single ordering of fabrication, exaggeration, and omission that holds across all datasets or depths.

Summary: A Representation-Dependent Problem. The results point to a consistent but nuanced picture of deception detection in LLMs. Depth improves separability in some datasets but not others, so later layers are not universally superior. More expressive probes help selectively, but standard logistic regression remains highly competitive and is often the most stable baseline. Sparse SAE features perform similarly to dense representations. Finally, lie typology clearly matters: fabrication, omission, exaggeration, and mixed training conditions produce different detection patterns depending on the dataset and representation layer. Overall, the experiments show that deception detection is best understood as a representation-dependent problem rather than a single-classifier problem.

6 Conclusion

This paper set out to examine deception detection in large language models through four factors: representation depth, probe expressivity, sparse representations, and lie typology. Across the experiments, the central conclusion is that deception is not encoded in a single uniform way. Instead, detectability depends heavily on the dataset context, the layer being probed, and the representation space used for classification.

For depth, the results show a dataset-dependent pattern rather than a universal rule. Some datasets benefit from deeper layers, while others remain more separable at earlier layers, meaning the commonly assumed "later is better" trajectory does not hold consistently across all deception settings. For probe expressivity, more complex models provide selective gains but do not dominate the simple logistic baseline. Truth2D offers the clearest structured improvement, while TPC, mass-mean, and INLP are useful in specific settings but not consistently stronger than a well-calibrated linear classifier. This suggests that deception detection is not solved simply by adding geometric or polynomial complexity.

For sparse SAE features, the results suggest that evaluating SAE representations at 66th percentile layer does not consistently enhance detectability over the original dense hidden states. In most settings, SAE preserves the core deceptive signal rather than improving it, and its benefits appear limited to a subset of harm-pressure cases. This suggests that, at least at this depth, SAE features function more as an alternative representation than as a reliably stronger one for deception detection. Finally, for lie typology, we found that fabrication, omission, exaggeration, and mixed training conditions do not behave uniformly. The strongest condition depends on both the dataset and the representation layer, indicating that the structural form of a lie materially affects its detectability, and no single lie type dominates universally.

The main challenge throughout this paper was that the deception signal was not stable across tasks; different datasets exposed different failure modes, and the optimal probing strategy frequently shifted with the evaluation context. Given that optimal detection performance depends heavily on highly localized factors—such as specific layers, lie typologies, and representation spaces—our findings strongly suggest that a monolithic detector is fundamentally limited. Instead, a highly promising avenue for future work lies in ensemble methods or a Mixture-of-Experts (MoE) probing framework. By training specialized, expert probes for distinct context types (e.g., separating safety-related choices from persona-driven roleplay) and utilizing a gating mechanism to dynamically route activations, such an architecture could explicitly leverage the selective strengths of the individual probes observed in our evaluation. Additionally, extending this analysis to the remaining Liars' Bench models would test whether these depth, sparsity, and typology patterns persist across different model families, while studying the SAE representations more directly could further bridge the gap between behavioral benchmarking and mechanistic interpretability.

Limitations

This study has several limitations. First, given the computational scope of a multi-factor study and the constraints of an extended abstract, we sampled two specific depth percentiles (20% and 66%)